Tianqi Xu

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Homepage | Google Scholar | LinkedIn | GitHub

Education

Sichuan University Chengdu, China
QS 2027: 16th in Mainland China; one of 39 universities selected for China's Project 985, a national initiative supporting leading universities.
B.Eng. in Software Engineering *Sep. 2023 – Jun. 2027 (Expected)*

- **GPA:** 3.78/4.0; **Weighted Average:** 89.82/100.
- **Advisor:** Tao He.

Research Profile

My research experience spans reasoning and generalization in Vision-Language-Action (VLA) models, test-time adaptation (TTA), and ODE-based representation learning for medical image segmentation. My long-term goal is to develop **lightweight methods for robotic self-improvement** that enable robots to learn efficiently from deployment experience and adapt to unfamiliar environments, supporting reliable real-world operation.

Research Interests

Vision-Language-Action Models; Self-Improving Embodied Models; Test-Time Adaptation; Efficient Reasoning.

Publications

1. **CNM-UNet: Continuous Ordinary Differential Equations for Medical Image Segmentation.**
   Tianqi Xu, Yashi Zhu, Quansong He, Yue Cao, Kaishen Wang, Zhang Yi, and Tao He.
   Proceedings of the AAAI Conference on Artificial Intelligence, **40**(14): 11406–11414, 2026. [DOI] [Paper] [Code]

Under Review

1. **SaCTTA: Selective Attention Continual Test-Time Adaptation for Medical Image Segmentation.**
   Tianqi Xu[*], Jun Wu[*], Junjie Hu, Jixiang Guo, Wei Liu, Lei Zhang, Zhang Yi, and Tao He.
   Under review at *Neurocomputing*. [*]Equal contribution.
2. **P-DORM: Parameter-Wise Dynamic Optimizer-State Reset for Test-Time Adaptation.**
   Tianqi Xu, Junjie Hu, Jixiang Guo, Haixian Zhang, Zhang Yi, and Tao He.
   Under review at *ICLR 2027*.

Research Experience

Reasoning and Generalization for VLA Mar. 2026 – Oct. 2026
University of Central Florida Orlando, FL, USA
Remote Research Intern; Advisor: Yuzhang Shang

- Investigated chain-of-thought reasoning for embodied decision-making and long-horizon task execution in VLA models.
- Formulated a conceptual RL strategy for VLA post-training, exploring reward signals derived from the model's reasoning and interaction history.

Stable Test-Time Adaptation via Optimizer-State Reset Feb. 2026 – Sep. 2026
Sichuan University Chengdu, China
Onsite Research Intern; Advisor: Tao He

- Developed a dynamic optimizer-state reset mechanism to address unstable updates during test-time adaptation.
- Designed and implemented parameter-aware thresholds and a directional alignment criterion for tensor-specific anomaly detection.

Selective Attention for Continual Test-Time Adaptation Aug. 2025 – Jan. 2026
Sichuan University Chengdu, China
Onsite Research Intern; Advisor: Tao He

- Developed a continual test-time adaptation method to address domain shifts in medical image segmentation.
- Implemented an attention-driven mechanism that treats layer outputs as information tokens to select layers for adaptation.
- Designed a self-distilled adaptive learning-rate strategy with sensitivity and consistency constraints to accelerate adaptation and mitigate error accumulation.

Continuous ODEs for Efficient Medical Image Segmentation Sep. 2024 – Jul. 2025
Sichuan University Chengdu, China
Onsite Research Intern; Advisor: Tao He

- Proposed CNM-UNet, a medical image segmentation model that replaces vanilla hierarchical decoders with a single Continuous Neural Memory ODEs block to minimize computational complexity.
- Leveraged Neural Memory ODEs for continuous temporal feature extraction, enabling precise data distribution modeling.
- Designed a Dual Self-updating strategy based on test-time adaptation principles to enhance cross-domain generalization.

Activities and Service

- **Presentations:** *CNM-UNet: Continuous Ordinary Differential Equations for Medical Image Segmentation.* Poster presentation, AAAI Conference on Artificial Intelligence (AAAI 2026), Singapore, Jan. 2026.
- **Academic Service:** Reviewer, International Conference on Learning Representations (ICLR 2027).
- **Student Activities:** Member, Publicity Department, Software Engineering Student Union, Sichuan University.
- **Community Service:** Volunteer, Wuhou Jifu Nursing Home, Chengdu, China.

Honors and Awards

- **Outstanding Graduate**, Sichuan University. 2026
- **Rising Star in Research Award**, College of Software Engineering, Sichuan University. 2026
- **Outstanding Student Award**, Sichuan University (three consecutive years). 2023–2026
- **Academic Excellence Scholarship**, awarded to the top 6% of students, Sichuan University. 2023–2024
- **Academic Excellence Scholarship**, awarded to the top 15% of students, Sichuan University. 2024–2025

Skills

- **Programming:** Python, Bash, C/C++, Java, SQL.
- **Tools & Frameworks:** PyTorch, Hugging Face, NumPy, pandas, OpenCV, Linux, Git, Docker, LaTeX.
- **Training & Inference:** SLURM-based GPU training (8 × NVIDIA H100); KV caching.
- **Languages:** English (TOEFL iBT: 4.5; CET-4: 585; CET-6: 560); Chinese (native).